Embedded Payments

Markets Potential of IOT Devices

What problem are we solving?

Powering seamless UPI payments though daily used IoT devices (cars, smart TVs, smart speakers etc.)

- The number of IoT devices are expected to reach 2.1 billion by 2025 in India and 19 billion worldwide
- IOT devices are expected to grow at a CAGR of 16% by 2030

Some Use cases of IOT based payments

Take me to nearest fuel station

Recharge my FasTag Wallet

On the Go 'Tap & Pay'

Hey Google Assistance
pay my Electricity Bill

- Source: Hash Studios Technologies (Click Here)
- Source: Exploding Topics (Click Here)
- Future Trends (Click Here)

IOT Payments With UPI

In today's era, people are buying more and more connected devices. IOT Payment solution aims to empower these devices with the capability of making payments with these devices itself.

Primary Device: Mobile Phone

User using UPI APP on mobile phone device to enable the IOT device for UPI Payments

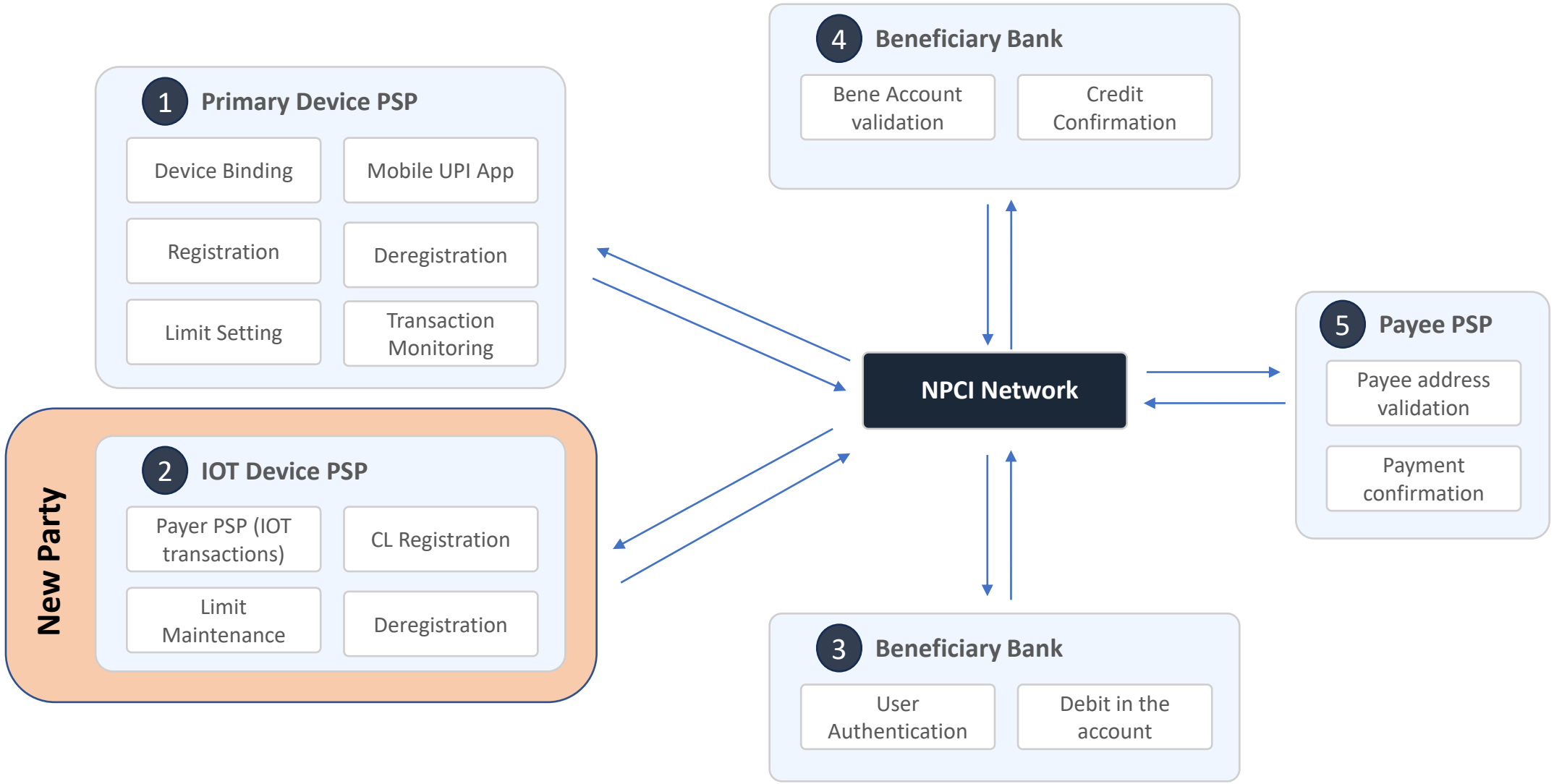
Secondary Device: IOT Device

User using IOT device to make IOT payments

Key Highlights

- Going to **5 Party Model** from current 4 Party Model
- Primary PSP and Secondary device PSP for payer to facilitate **Interoperable Registration** and **Remote Lifecycle Management**
- For payment and authentication, transactions will be initiated from IOT device.
- Limit setting through UPI Mobile App,

Architecture diagram



Risk Mitigation

1. **Limit Setting:** Limits for the dependent device will be set through the primary app to limit exposure

2. **Deregistration**: allowed from the primary app and secondary app for the dependent device in case user is not using the device

3. **Cooling Period**: During Cooling Period of first 24 hours, reduced limits will be levied, similar to existing UPI guidelines

4. **Per transaction and daily cumulative limit** will be set at NPCI based on the usecase.

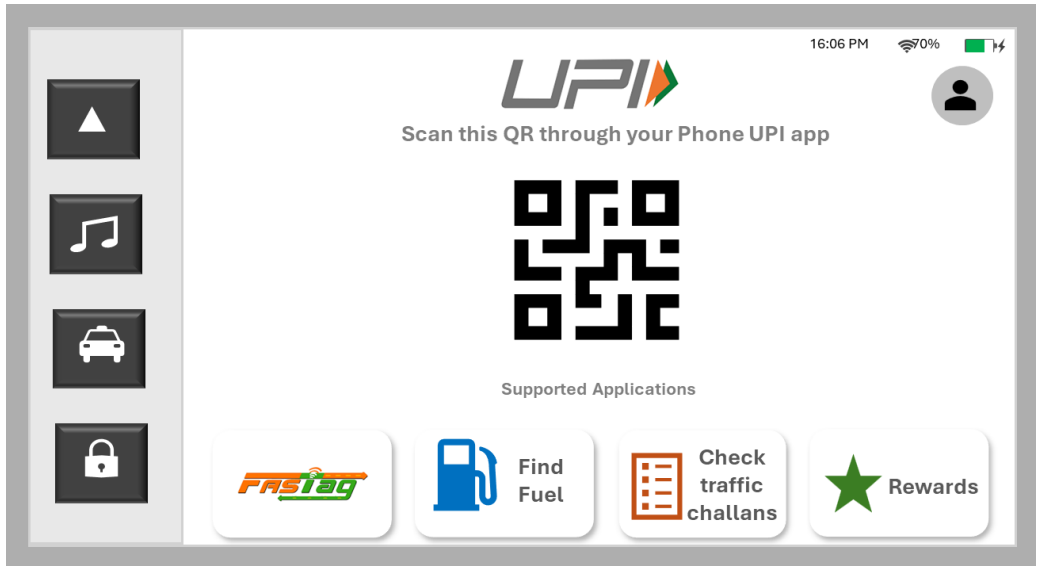
Indicative User Journey

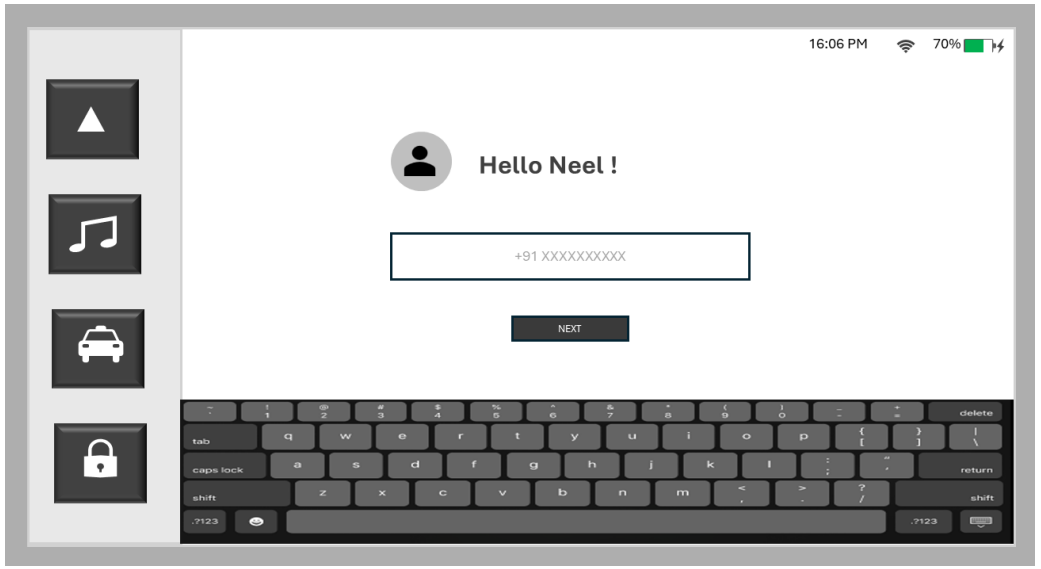
NPCI
भारतीय राष्ट्रीय भुगतान निगम
NATIONAL PAYMENTS CORPORATION OF INDIA

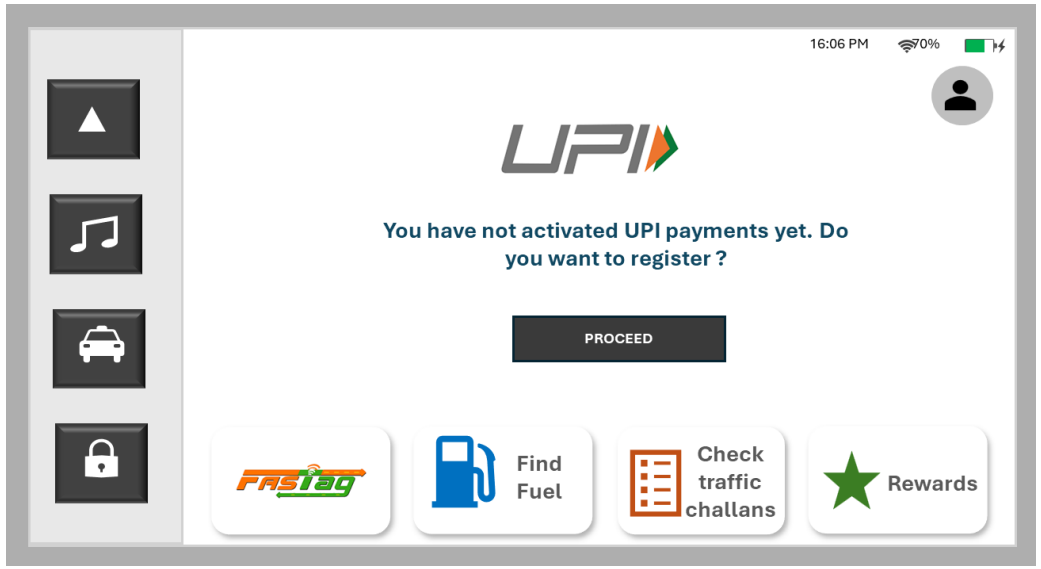
Registration QR on IOT Device

Registration QR is generated on IOT device

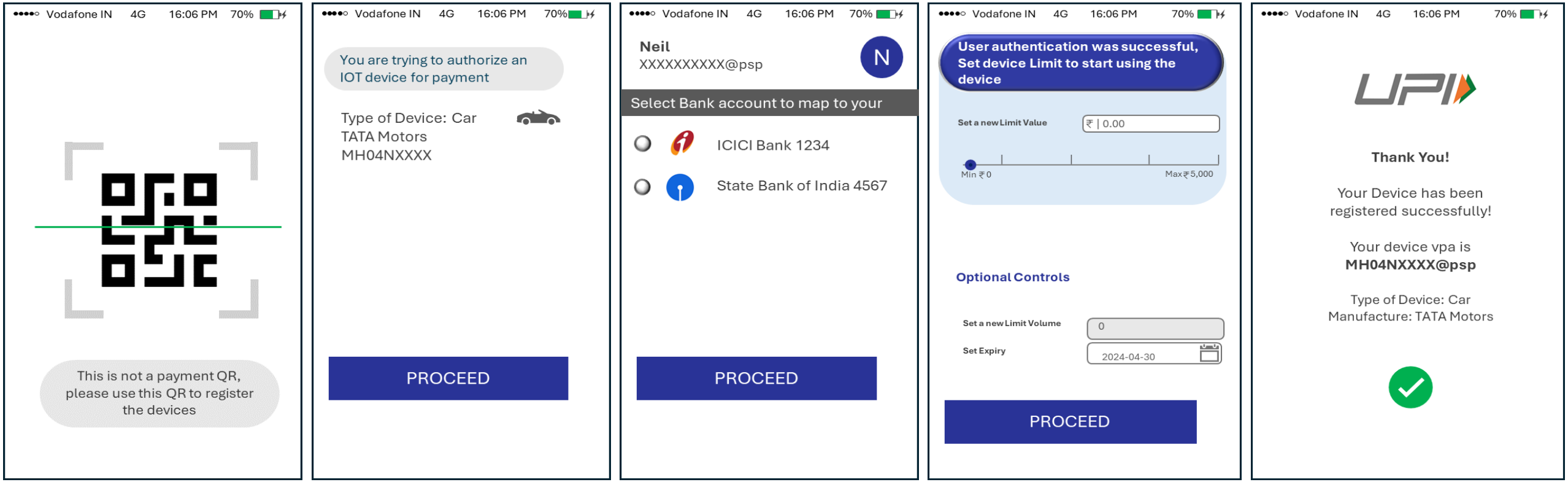
User will scan this QR to register from Mobile UPI APP





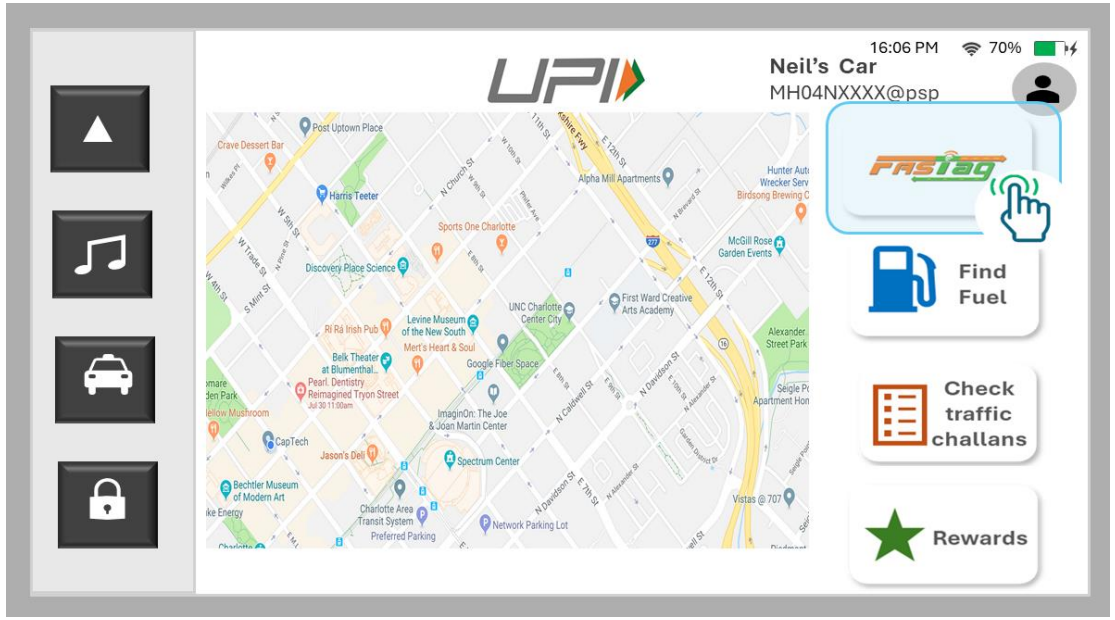


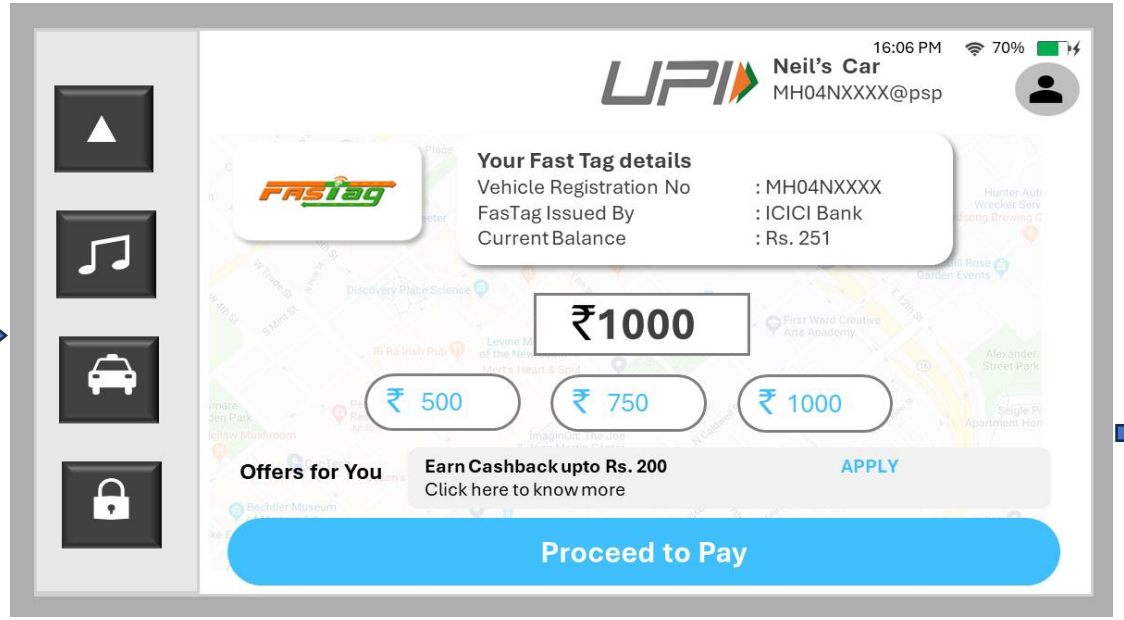
IOT Device Registration from UPI app in Primary Device

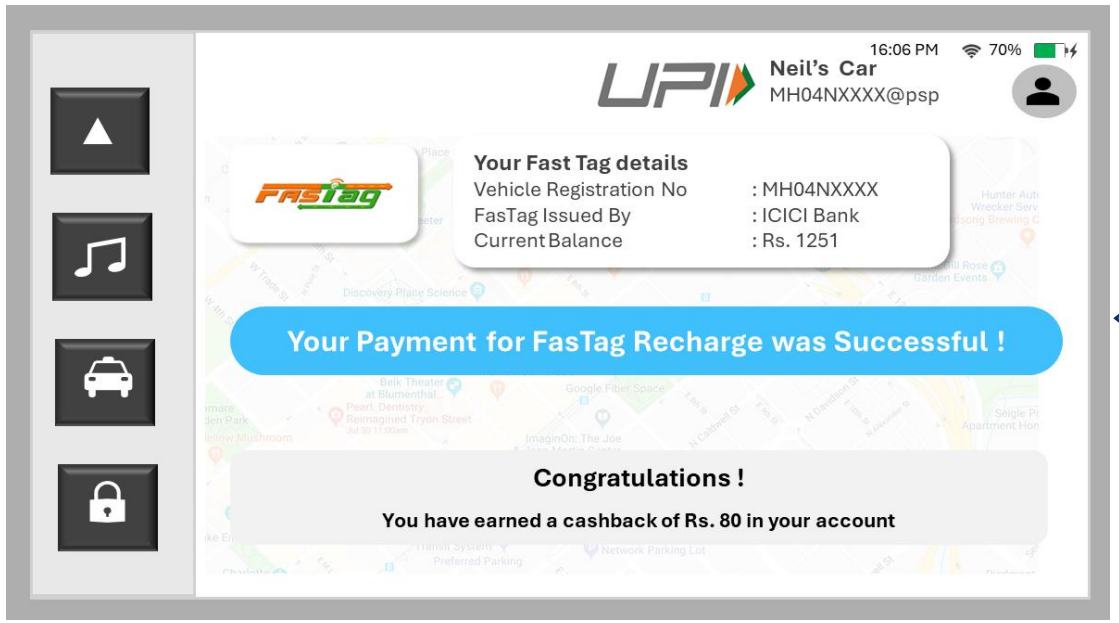


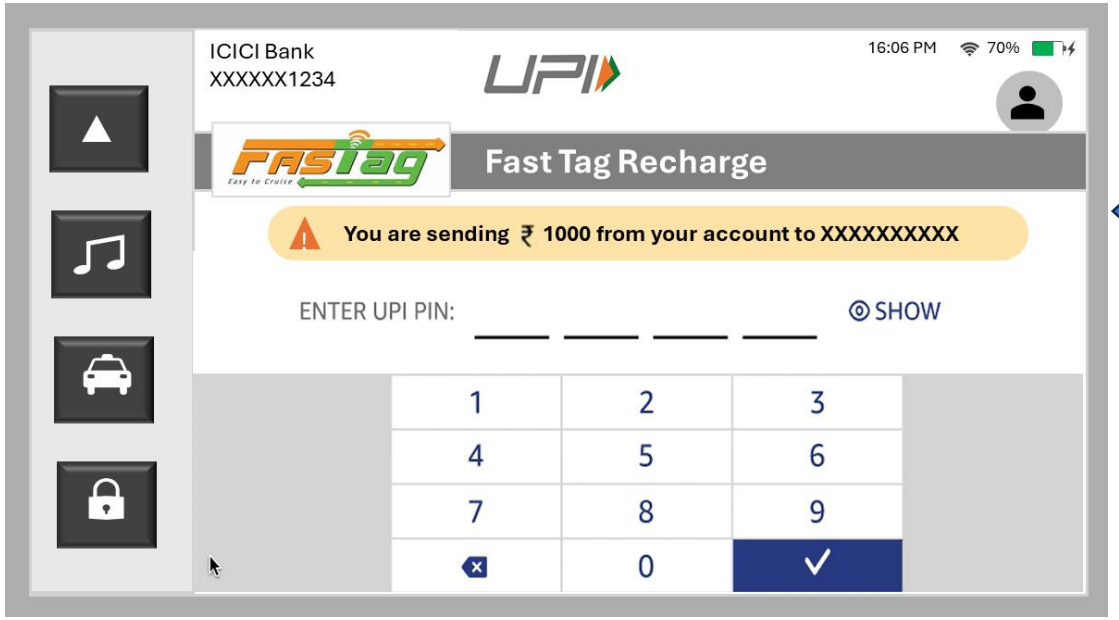
1. Scan QR from IOT device
2. Confirm Device Details
3. Confirm Banking details
4. Define Limit
5. Registration Confirmation

Payment from IOT Device: Car Infotainment

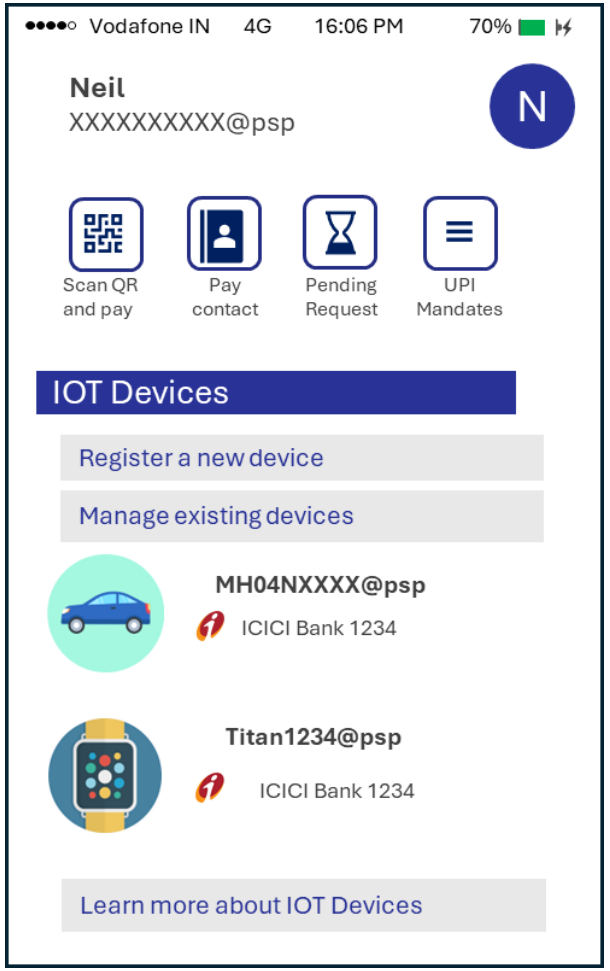


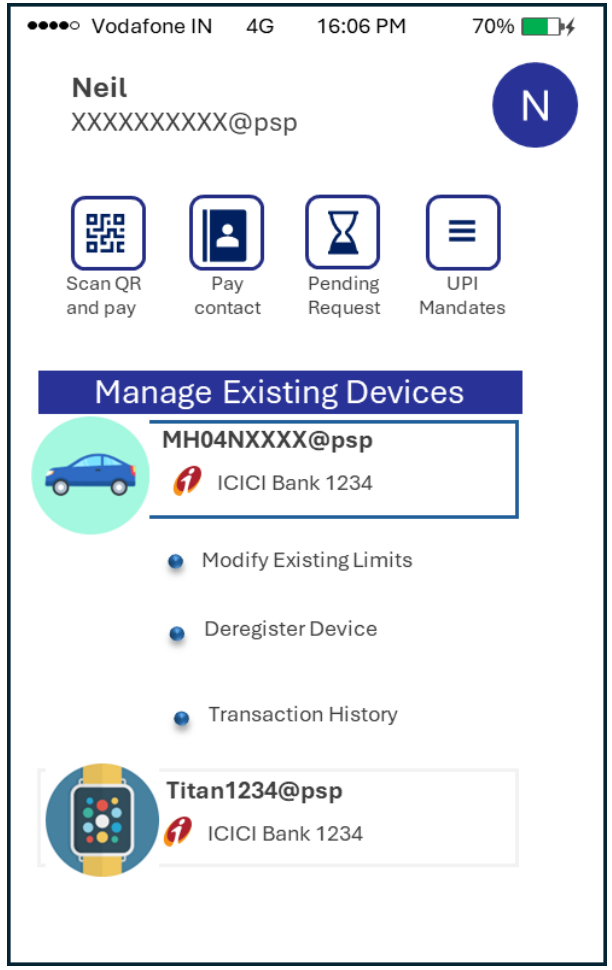


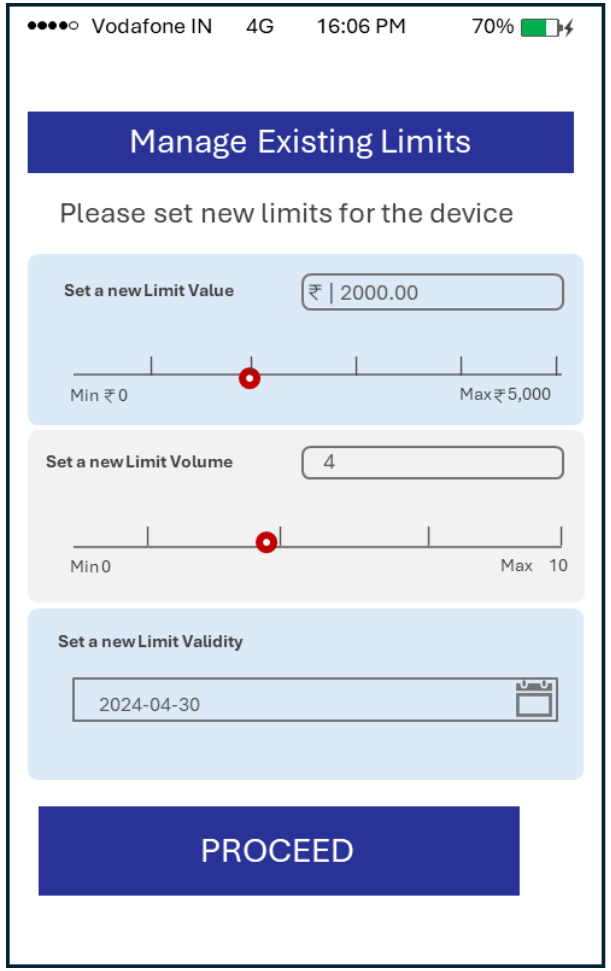


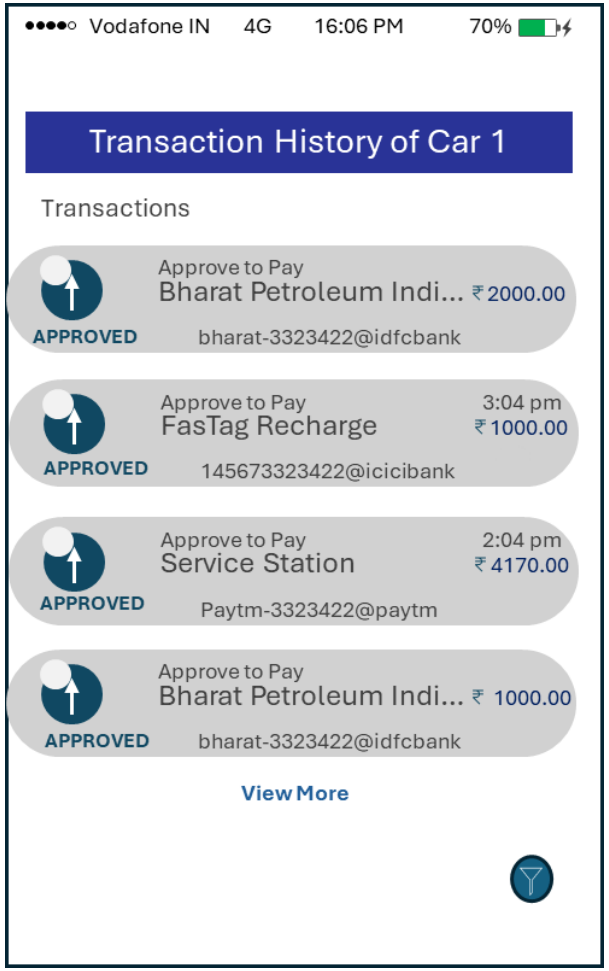


IOT Device Management from UPI app in Mobile



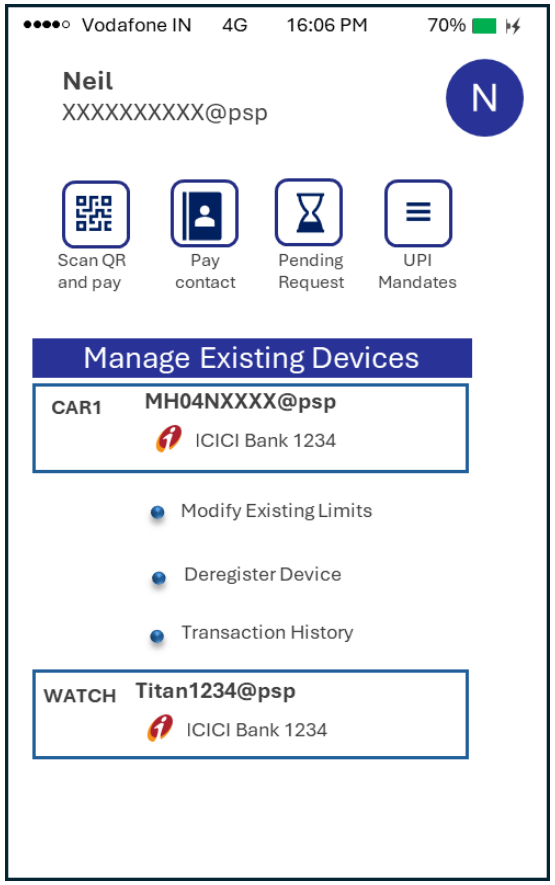


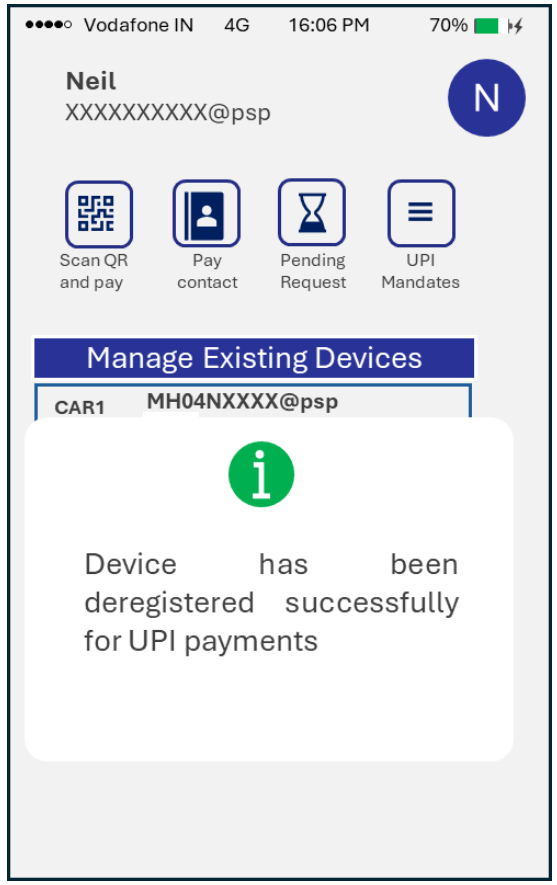




Deregistration of IOT Devices

Option1: Deregistration from Primary Device





Option2: Deregistration from IOT Device